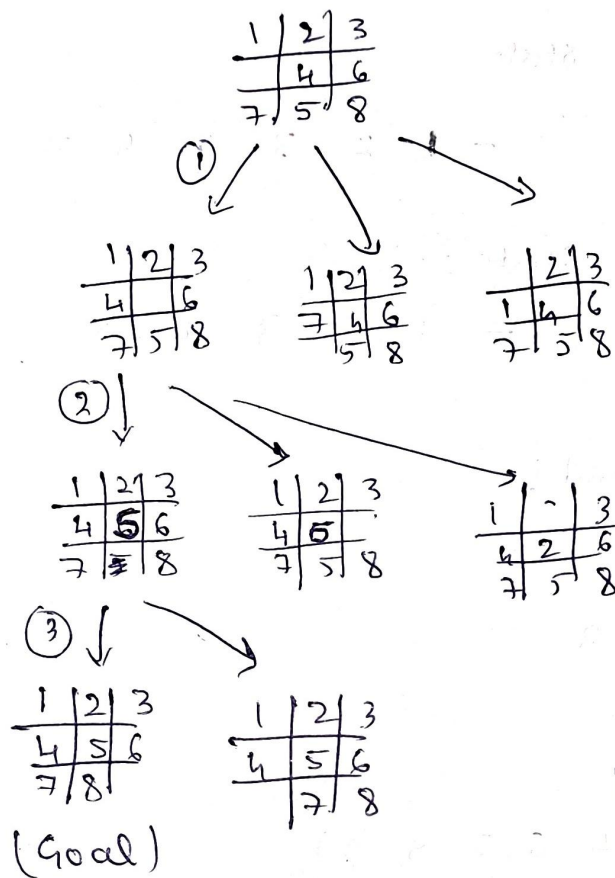


Q1. 8-puzzle (IDS)



path cost = 3

Pseudocode / Algorithm

> IDS (start, goal, maxDepth)

1. for depth = 0 to maxDepth
 - a. call DLS (start, goal, depth)
 - b. If solution found, return it.
2. Return "No solution"

DLS (start, goal, depth)

1. If state = goal, return "success"
2. If depth = 0, return "failure"
3. Generate all valid child states.

for each child:

call DLS (child, goal, depth-1)

if success:

return "success"

return "Failure"

Output:

Enter initial state :

- 1 2 3 0 4 7 5 8

Enter goal state :

- 1 2 3 4 5 6 7 8 0

Solution Found!

Depth : 3

Path cost : 3

Path:

(1, 2, 3, 0, 4, 6, 7, 5, 8)

(loop)

(1, 2, 3, 4, 0, 6, 7, 5, 8)

initial loop / absolute

(1, 2, 3, 4, 5, 6, 7, 0, 8) (loop, loop, loop) and

(1, 2, 3, 4, 5, 6, 7, 8, 0)

0 = goal of

(loop, loop, loop) and

to reach goal, regular

regular

(loop, loop, loop)

"loop" and

loop, loop, loop

loop, loop, loop

loop, loop, loop

loop, loop, loop

loop, loop, loop

22. 8 puzzle (using A* Algorithm).

A* (start, goal)

OPEN = { start }

while OPEN is not empty:

current = node in OPEN with lowest $f(n)$

if current == goal:

return path to goal

remove current from OPEN

for each neighbor of current:

calculate new path cost $g(\text{neighbor})$

if this path is better:

update parent of neighbor

update $g(\text{neighbor})$

$f(\text{neighbor}) = g(\text{neighbor}) + h(\text{neighbor})$

add neighbor to OPEN if necessary

return "failure"

i/p: Enter initial state : 2 8 3 1 6 4 0 7 5

Enter goal state : 1 2 3 8 0 4 7 6 5

Optimal solution found in 6 moves!

Step 0

2 8 3 1 6 4 0 7 5

$g(n) = 0$ $h(n) = 6$ $f(n) = 6$

Step 1

2 8 3 1 6 4 7 0 5

$g(n) = 1$ $h(n) = 5$ $f(n) = 6$

Step 2

2 8 3 1 0 4 7 6 5

$g(n) = 2$ $h(n) = 4$ $f(n) = 6$

Step 3

2 0 3 1 8 4 7 6 5

$g(n) = 3$ $h(n) = 3$ $f(n) = 6$

$$g(n) = 0$$

2	8	3
1	6	4
7	7	5

1	2	3
8		4
7	6	5

$$g(n) = 1$$

2	8	3
1		4
7	6	5

$$h(n) =$$

2	8	3
1	6	4
	7	5

$$h(n) =$$

2	8	3
1	6	4
7	5	

$$h(n) =$$

$$g(n) = 0$$

2	8	3
1	6	4
	7	5

1	2	3
8		4
7	6	5

$$f(n) = h(n) + g(n)$$

$$g(n) = 1$$

2	8	3
	6	4
1	7	5

$$h(n) = 5$$

$$f(n) = 6$$

2	8	3
1	6	4
7		5

$$h(n) = 4$$

$$f(n) = 5$$

$$g(n) = 2$$

2	8	3
1		4
7	6	5

$$h(n) = 3$$

$$f(n) = 5$$

2	8	3
1	6	4
7	5	

$$h(n) = 5$$

$$f(n) = 7$$

2	8	3
1	6	4
7	5	

$$h(n) = 5$$

$$f(n) = 7$$

$$g(n) = 3$$

2	8	3
1		4
7	6	5

$$h(n) = 3$$

$$f(n) = 6$$

2		3
1	8	4
7	6	5

$$h(n) = 3$$

$$f(n) = 6$$

2	8	3
1	6	4
7		5

$$h(n) = 4$$

$$f(n) = 7$$

2	8	3
1	4	
7	6	5

$$h(n) = 4$$

$$f(n) = 7$$

$$g(n) = 4$$

2	8	3
1	6	5
7		

2	8	3
7	1	4
	6	5

1	2	3
1	8	4
7	6	5

2		3
1	8	4
7	6	5

$$g(n) = 4$$

1	2	3
1	8	4
7	6	5

$$h(n) = 2$$

$$f(n) = 6$$



$$g(n) = 5$$

1	2	3
	8	4
7	6	5

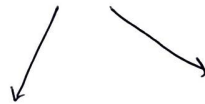
$$h(n) = 1$$

$$f(n) = 6$$

2	1	3
1	8	4
7	6	5

$$h(n) = 3$$

$$f(n) = 8$$



$$g(n) = 6$$

1	2	3
8		4
7	6	5

$$h(n) = 0$$

$$f(n) = 6$$

1	2	3
7	8	4
	6	5

$$h(n) = 2$$

$$f(n) = 8$$

—(Goal)—

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(op continued)

Step 4

0 2 3 1 8 4 7 6 5

$$g(n) = 4 \quad h(n) = 2 \quad f(n) = 6$$

Step 5

1 2 3 0 8 4 7 6 5

$$g(n) = 5 \quad h(n) = 2 - 1 = 1 \quad f(n) = 6$$

Step 6

1 2 3 8 0 4 7 6 5

$$g(n) = 6 \quad h(n) = 0 \quad f(n) = 6$$